Toward Symbolic-Aware Music Language Models

Compact Learned Symbolic Planning Tokens for Controlled Music Fusion

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Audio Tokens Do Not Reliably Encode Musical Structure

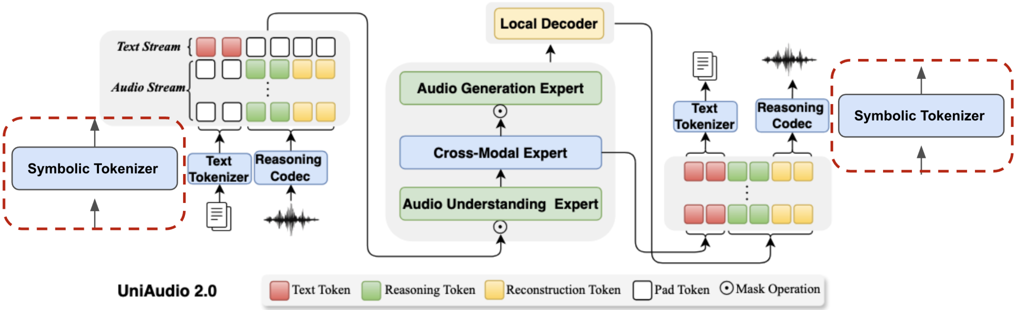
What audio tokens can encode	What they miss
Timbre, loudness, texture	Key signature, chord progression
General instrument colour	Stem-level instrument identity
Rough tempo feel	Exact meter and beat phase
High-level scene	Phrase boundaries, section structure

Central question: can MIDI-like symbolic structure be injected into a LALM in a controlled, verifiable way?

Three Research Questions Organize This Work

Question	What we need to show
RQ1 Music-aware understanding	Does aligned symbolic structure improve music-aware understanding? Metric evidence: aligned CE $<$ shuffled/dummy controls.
RQ2 Alignment-specific understanding	Does the benefit depend on correct symbolic-audio alignment? Metric evidence: aligned CE $<$ shuffled/dummy; gate higher.
RQ3 Music-structural understanding	Does S_{plan} support music-structural understanding? Metric evidence: MIR probing gaps over zero-feature baselines.

Our Approach: S_{plan} as a Third Stream in UniAudio 2.0

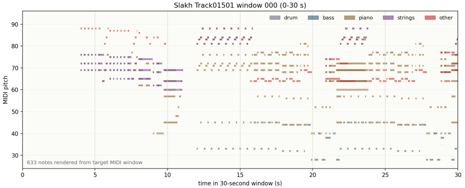


Dashed branch = symbolic tokenizer. In our implementation, this is S_{plan}: MIDI features $\rightarrow$ compact RVQ tokens inserted between R_a and C_a.

Dataset: Slakh2100 — Synthesized Paired Audio-MIDI

Split	Tracks	Windows
Train	1,289	11,279
Validation	270	2,334
Test	151	1,385

- Synthesized with professional-grade sample-based virtual instruments
- Track-level split *before* windowing — no phrase contamination
- Each window: 30 s audio + ground-truth aligned MIDI
- All CE results in this talk are on the held-out **test** set



Example 30 s piano roll — Slakh track 01501

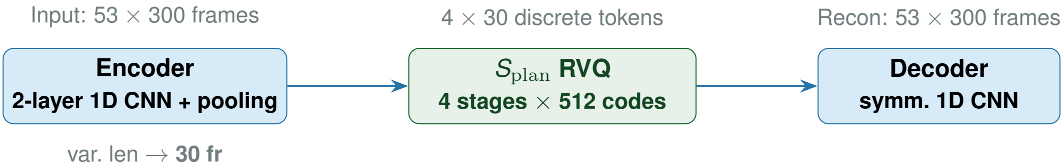
Method: MIDI Feature Extraction — 53-Dim Frame Vector

Extracted at 100 ms resolution from ground-truth MIDI

Feature group	Dims	Content
Binary features	41	Instrument-class presence (34 Slakh classes), pitch-class activation (12), beat/downbeat phase
Continuous features	12	Instantaneous tempo (BPM), note density (events/s), polyphony count, energy proxy
Frame vector	**53**	Normalized per-dimension; $\approx$300 frames per 30 s window

Oracle MIDI features; codec compresses $\approx$ 300 frames to **30**.

Method: S_{plan} RVQ Codec Architecture



Encoder	2-layer 1D CNN + temporal pooling; maps variable-length input to 30 frames
Codebook	4 RVQ stages, 512 codes each; EMA updates + codebook restart
Training loss	BCE for binary dims + MSE for continuous dims; 42,000 steps

$4 \times 30 = 120$ tokens per window. Same RVQ format as C_a — stream-native positional encoding.

Experiment Setup: LoRA Fine-tuning with Learned Gate

Component	Configuration
Base model	UniAudio 2.0 (frozen weights)
Adapters	LoRA on 140 attention modules
Trainable params	$\sim$10.8M (reason-only) $\sim$253M (symbolic variants)
Symbolic gate	Learned scalar; controls S_{plan} influence on C_a prediction
Hardened four-condition comparison (primary result)	
Steps / batch	800 steps; batch 1 + gradient accumulation 16
Learning rates	LoRA 10^{-5} / symbolic 3×10^{-4} / gate 2×10^{-3}
Seed	Fixed; all four conditions identical except stream variant

Gate $\approx$ 0: model ignores S_{plan}. Gate $>$ 0: symbolic stream admitted to C_a prediction. Primary result: aligned 0.177 $>$ shuffled 0.167 $>$ dummy 0.163.

Evaluation Protocol: Every Comparison Uses All Four Conditions

Condition	**Content of S_{plan} slot**
Aligned S_{plan}	Tokens from the *same* window's MIDI
Shuffled S_{plan}	Tokens from a *different* window's MIDI
Dummy stream	Zero / random RVQ codes
Reason-only	No S_{plan} slot at all

RQ1: Does S_{plan} improve music-aware understanding?

CE proxy: frozen codec + proxy head; 5,000 steps

Condition	Valid CE	Test CE	Test PPL
Reason only	7.565	7.657	2116
Low-rate rule summary	7.553	7.645	2089
Aligned S_{plan}	**7.507**	**7.596**	**1991**
$\hookrightarrow$ Shuffled S_{plan}	7.538	7.631	2061
$\hookrightarrow$ Dummy stream	7.602	7.694	2196
$\hookrightarrow$ Random stream	7.540	7.632	2064

Initial CE evidence: aligned S_{plan} gives the strongest gain.

RQ1: Does S_{plan} improve music-aware understanding?

CE evidence in stream-native $R_a \to S_{\mathrm{plan}} \to C_a$ layout

Condition	Test CE	Test Acc	Gate
Aligned S_{plan}	**5.766**	**0.1097**	0.003
$\hookrightarrow$ Shuffled	5.785	0.1067	0.003
$\hookrightarrow$ Dummy	5.785	0.1068	0.003

Initial CE evidence replicates in stream-native integration.

RQ1: Does S_{plan} improve music-aware understanding?

Primary CE result: hardened four-condition comparison

Condition	Test CE	Test PPL	Acc	Gate
Reason-only baseline	5.859	350	0.104	0.000
Aligned S_{plan}	**5.600**	**270**	**0.121**	**0.177**
$\hookrightarrow$ Shuffled S_{plan}	5.614	274	0.119	0.167
$\hookrightarrow$ Dummy stream	5.614	274	0.119	0.163

RQ1: positive initial evidence. Metric: lowest CE under matched controls.

RQ2: Is music-aware understanding alignment-specific?

CE/gate evidence: capacity effect vs. aligned-content effect

Effect	Δ CE
Stream capacity (any symbolic stream vs. reason-only)	−0.245
Alignment-specific (aligned vs. both controls)	−0.014
Total (aligned vs. reason-only)	**−0.259**

Comparison	Steps	Aligned vs. shuffled
Proxy task	5,000	−0.034
Stream-native warmup	3,000	−0.018
Hardened comparison	800	−0.014

RQ2: positive initial evidence. Metrics: aligned CE below controls; gate higher.

RQ3: Does S_{plan} support music-structural understanding?

Metric evidence: MIR probing with a zero-feature baseline

MIR attribute	Aligned S_{plan}	Zero-feature control
Weak key accuracy	**0.557**	0.126
Pitch-class macro-F1	**0.854**	0.830
Instrument macro-F1	**0.501**	0.426
Segment density acc.	0.874	0.879
Instrument micro-F1	0.791	0.835
Meter accuracy	0.783	0.855

RQ3: qualified positive. Metrics: strongest probing gaps for key and pitch-class.

RQ3: Does S_{plan} support music-structural understanding?

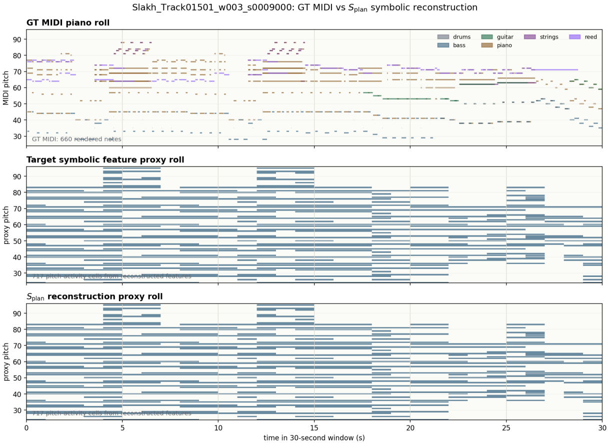
Metric evidence: supported axes and prior-dominated axes

Axis	Gap	Status
Weak key accuracy	$+0.431$	**Strongest support.**
Pitch-class macro-F1	$+0.024$	Positive feature-specific signal.
Instrument macro-F1	$+0.075$	Positive, with class-imbalance sensitivity.
Instrument micro-F1	-0.044	Prior-dominated. Common-class baseline wins.
Meter accuracy	-0.072	Prior-dominated. Common-meter baseline wins.

Gap = aligned S_{plan} score - zero-feature control.

RQ3: Does S_{plan} support music-structural understanding?

Qualitative reconstruction: Slakh Track01501, window 3 (90–120 s)



Current Scope and Next Steps

Gap	Why it matters
No decoded generation	CE results are on semantic tokens only. No waveform decoded from aligned vs. shuffled.
Oracle MIDI input	Aligned MIDI unavailable at inference on real audio. Needs audio-to-S_{plan} predictor.
No LLM-supervised tokenizer	Text-language grounding (open QA, captioning) not yet trained.
MIR controls in progress	Label-shuffle and target-prior controls will strengthen the probing analysis.
Slakh only	No real-recording or cross-domain transfer evidence.

Next priority: decode C_a from the aligned checkpoint to waveform and compare perceptually against shuffled — connecting the token-level result to perceptual audio evidence.

Conclusion: Three Research Questions Answered

RQ	Finding
RQ1	**Music-aware understanding — Positive initial evidence.** CE improves from 5.859 to 5.600 under matched controls.
RQ2	**Alignment-specific understanding — Positive initial evidence.** CE: aligned beats controls; gate $0.177 > 0.167 > 0.163$.
RQ3	**Music-structural understanding — Qualified positive.** Probing gaps support key ($+0.431$) and pitch-class ($+0.024$).

Takeaway: controlled baselines make symbolic-fusion evidence interpretable.

Backup: Why Not Just Append Symbolic Tokens?

Naive side-channel insertion does not use symbolic content

Side-channel insertion:

Condition	**Test CE**	**vs. reason-only**
Reason-only	**5.360**	baseline
Aligned S_{plan}	5.511	$+0.151$
Shuffled S_{plan}	5.512	$+0.152$
Dummy	5.510	$+0.150$

Takeaway: S_{plan} must enter through the stream-native $R_a \to S_{\mathrm{plan}} \to C_a$ interface.

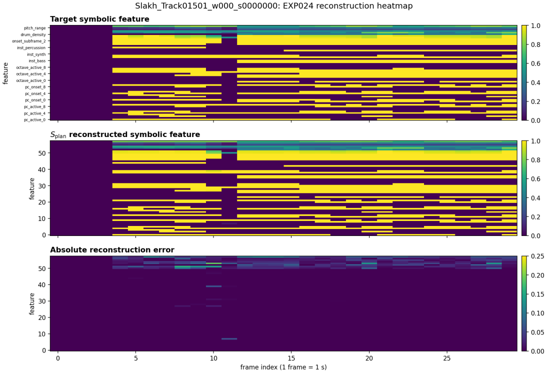
Backup: Symbolic Codec Reconstruction

Metric	Value
Validation reconstruction loss	0.00918
Binary feature accuracy	**0.99777**
Continuous feature MSE	**0.00172**
Codebook utilization — books 1/2/3/4	31% / 72% / 81% / 81%
Fixed window `w000` binary acc / MSE	1.000 / 0.00174
Fixed window `w001` binary acc / MSE	1.000 / 0.00246
Fixed window `w003` binary acc / MSE	1.000 / 0.00130

Codec check: faithful reconstruction without codebook collapse.

Backup: Feature Heatmap — Target vs. Reconstruction

Slakh Track01501, window 0 (0–30 s) — binary acc 1.000, MSE 0.00174



Backup: Per-Codebook Cross-Entropy

Symbolic benefit distributed across all 8 acoustic codebooks (hardened comparison)

Condition	**cb0**	**cb1**	**cb2**	**cb3**	**cb4**	**cb5**	**cb6**	**cb7**
Reason-only	5.60	6.00	3.45	5.28	6.02	6.48	6.87	7.16
Aligned S_{plan}	**5.24**	**5.71**	**3.19**	**4.98**	**5.75**	**6.25**	**6.68**	**7.01**
Shuffled	5.26	5.73	3.20	4.99	5.76	6.26	6.69	7.02
Dummy	5.26	5.73	3.20	4.99	5.76	6.26	6.69	7.02

Symbolic benefit is distributed across all 8 codebooks. Books 0, 2, and 3 show the largest gains, suggesting symbolic structure acts as a long-range prior that helps coarser-grained acoustic prediction first.

Backup: Per-Window Case Spectrum (Stream-Native Warmup)

Mixed per-window margins in the stream-native warmup

Window	Aligned CE	Best control	Margin	Note
`w000`	5.226	5.280	$+0.054$	Strong positive
`w001`	6.159	6.163	$+0.004$	Small positive
`w002`	6.097	6.091	-0.006	Slight negative
`w003`	5.876	5.869	-0.007	Slight negative

Aggregate positive result holds even though individual windows show mixed margins. The warmup contains both strong positive and near-zero examples.